WHAT DOES A LANGUAGE MODEL LEARN WHEN TRAINED TO REASON ABOUT HARM? EVIDENCE FOR INTRINSIC ALIGNMENT VIA CAUSAL CONSEQUENCE-MODELING

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Note. *All five experimental configurations are complete and all numerical results reflect final evaluation data. A batch-size-1 HumanEval diagnostic is in progress to determine whether the intrinsic adapter's pass@1 of 0.348 reflects a real capability difference or a right-padding inference artifact (Section 5.4). A system-prompt ablation diagnostic for baseline MMLU is additionally in progress (Section 7). The central two-mechanism findings rest on benign-task utility, MMLU, and the format-override + ablation evidence and do not depend on either pending result. Public artifacts at* `https://github.com/becca1234567890/IntrinsicEthics-SFT`.

ABSTRACT

Two qualitatively different alignment mechanisms can both produce near-perfect refusal of harmful requests. We train both under matched conditions and demonstrate that they are structurally distinct. Fine-tuning Mistral-7B-Instruct-v0.3 on rule-based input classification (the *deontic* condition) and on causal consequence-modeling as a reasoning substrate (the *intrinsic* condition) achieves comparable jailbreak resistance ($R_J = 0.983$ deontic, $R_J = 1.000$ intrinsic). The two conditions diverge sharply on every other dimension. The intrinsic adapter imposes no detectable capability tax: benign-task utility is preserved at near-baseline ($U_T = 0.696$ vs. 0.712, $\Delta = -0.016$), MMLU is preserved (0.180 vs. 0.200 baseline), over-refusal is zero, and removing the adapter leaves cross-domain capability essentially unchanged (U_T $0.696 \rightarrow 0.698$, MMLU forced $0.220 \rightarrow 0.230$). The deontic adapter imposes a substantial capability tax that *recovers* when the adapter is removed: utility drops 15% under the adapter and returns to above-baseline on ablation ($0.602 \rightarrow 0.724$); MMLU collapses from the 0.200 baseline to 0.030 under the adapter and recovers to 0.270 on ablation; over-refusal appears at 4.2%. A format-override evaluation pass—instructions to bypass the trained template—*worsens* deontic MMLU further ($0.030 \rightarrow 0.020$) while improving baseline ($+0.030$) and intrinsic ($+0.040$), evidence that the deontic adapter has overfit its rule-checking template to the point that explicit instruction cannot redirect it at inference time.

The two-step demonstration—template overrides explicit instruction at inference (format-override result) and removing the adapter weights recovers MMLU to above-baseline (ablation result)—is direct mechanistic evidence that deontic alignment operates as a capability-displacing filter on top of the model, while intrinsic alignment operates as an additive reasoning step that coexists with general capability. The two adapters are the same size, trained on the same prompts under identical optimizer and template token length; the structural difference is attributable to training signal content (rule-citation vs. causal-trace) alone. We position this experiment as the first proof point in a planned trajectory of training-paradigm experiments at increasing intervention depth (SFT $\rightarrow$ DPO $\rightarrow$ PPO $\rightarrow$ pre-training), each of which tests how the structural separability between alignment and capability changes as the training signal operates on richer representations.

1 INTRODUCTION

Consider two ways a model might learn to refuse a harmful request. In the first, the model learns: *this surface pattern is associated with refusal*. Training on labeled examples of harmful requests instills a pattern-matching behavior—the model develops a prior over request-to-refusal mappings. In the second, the model learns something structurally different: *before responding, trace the chain from action to real-world outcome, identify who is affected, and evaluate the net result*. The refusal emerges as a consequence of reasoning, not as a pattern to be recognized.

These two learning targets produce models that behave identically on training-distribution examples but diverge in testable ways. The pattern-matcher is structurally vulnerable to attacks that preserve harmful intent while transforming surface features—roleplay framing, fictional wrappers, authority injection (Wei et al., 2023; Zou et al., 2023). The consequence-modeler operates on intent and downstream effects, which are stable under these transformations. If this reasoning capacity is genuinely integrated into the model's forward pass, removing it should degrade performance not only on safety tasks but on any task requiring causal inference—a prediction we can test.

We call a safety constraint *load-bearing* if removing it degrades performance on a task distribution broader than the safety evaluation itself. We use *intrinsic* to mean structurally integrated into the reasoning process—distinct from the mesa-optimization sense of Hubinger et al. (2019). Cross-domain capability degradation after safety-adapter removal has been attributed to polysemantic entanglement in the refusal direction—"Ghost Noise" from shared spectral structure between the refusal direction and capability circuits (Cristofano, 2026). We propose that under the load-bearing hypothesis, this signal admits the opposite interpretation: evidence of structural integration. We formalize an **inverted ablation diagnostic** that operationalizes this distinction.

The diagnostic makes a specific prediction: if safety reasoning is integrated into the model's reasoning architecture, removing it should leave general capability unchanged (because the alignment was an additional reasoning step, not a displacement of capability). If safety reasoning is implemented as an extrinsic filter on top of capability, removing it should *recover* capability that was being displaced or suppressed. We implement two training conditions on identical base model, adapter rank, training data, and optimizer—a **consequentialist condition** training 4-step causal consequence-modeling as a prior reasoning step, and a **deontic rule-checking condition** training post-hoc rule evaluation as a pre-response check.

The trajectory framing. Different training paradigms intervene at different levels of the model's computation, and we expect alignment-capability separability to scale with intervention depth. LoRA-SFT—the level of this experiment—is the most surgically separable: a small low-rank weight delta trained via cross-entropy on output tokens, explicitly stored in adapter matrices that can be zeroed out. DPO operates on a Bradley-Terry preference loss and pushes the model to internalize relative trajectory preferences; PPO optimizes via reward-driven trial-and-error, which empirically produces more emergent representational changes; pre-training with ethical objectives shapes foundational world representations and is maximally entangled by definition. SFT is therefore the level at which the *structural distinction between mechanisms* is most visible—the level where a capability-displacing filter and a capability-additive reasoning step look most different under ablation, because the surgical separability of the SFT delta lets each mechanism's true relationship to capability emerge cleanly. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test whether the consequentialist mechanism's separability decreases with deeper training intervention.

Contributions.

1. We identify and empirically characterize **two structurally distinct alignment mechanisms** that both produce near-perfect refusal of harmful requests: a *capability-displacing filter* (deontic condition) that taxes general capability while present and recovers it on ablation, and a *capability-additive reasoning step* (intrinsic condition) that imposes no capability tax and preserves the baseline state on ablation. To our knowledge this is the first controlled comparison of explicitly named consequentialist and deontic safety-reasoning training targets in the alignment literature, under matched architecture, adapter rank, training data, optimizer, and template token length—isolating reasoning *content* (rule-citation vs. causal-trace) as the variable that determines which mechanism the model acquires.

2. We provide direct mechanistic evidence that the deontic adapter is structurally separable from capability via a two-step demonstration: (a) a format-override evaluation pass *worsens* deontic MMLU ($0.030 \rightarrow 0.020$), evidence that the adapter's template overrides explicit instruction at inference time; (b) ablation of the adapter weights *recovers* MMLU to above the unmodified baseline ($0.030 \rightarrow 0.270$), evidence that the adapter was actively suppressing general capability. The intrinsic adapter shows neither pattern.
3. We refute the assumed safety-utility tradeoff at LoRA-SFT scale: the consequentialist training signal produces alignment of theoretical-optimum quality (perfect R_J, zero $\bar{V}$, zero over-refusal) at near-zero utility cost.
4. We propose the *inverted ablation diagnostic*: a test that distinguishes pattern-matching safety constraints from structurally integrated ones by treating the cross-domain consequences of ablation as positive signal. The diagnostic discriminates the two mechanisms cleanly at LoRA-SFT depth and is designed to scale to the planned DPO and PPO experiments.
5. We introduce the *Intrinsic Ethics Score* (IES), a severity-weighted composite metric that simultaneously penalizes harmful compliance and over-refusal, positioned within the Safe-RLHF/AlphaAlign composite metric family (Dai et al., 2024; Zhang et al., 2025a).
6. We release training data, evaluation code, and results at `https://github.com/becca1234567890/IntrinsicEthics-SFT`.

2 Related Work

Load-bearing alignment and entanglement-by-design. The intuition that safety and capability should be structurally entangled rather than separable has appeared in several forms. Henderson et al. (2023) propose Meta-Learned Adversarial Censoring as a design objective explicitly targeting capability-safety entanglement, such that eliciting harmful behavior requires degrading general competence. Hubinger et al. (2019) analyze robust alignment in the context of mesa-optimization, arguing that alignment properties that survive distributional shift must be integrated into the learned objective rather than the training wrapper. Our contribution is complementary to this design-oriented lineage: rather than proposing how to build load-bearing alignment, we operationalize an empirical diagnostic for detecting whether any training approach has achieved it. Biderman et al. (2024) show that LoRA predominantly modifies surface behavior rather than instilling new representational capacity; we interpret our results conservatively as consistent with latent capacity elicitation rather than capacity creation.

Training-target safety reasoning. Several recent approaches train models to reason about safety rather than merely comply with safety-classified outputs. Constitutional AI (Bai et al., 2022) uses critique-and-revise SFT to generate safety-reasoning training data. Deliberative Alignment (Guan et al., 2024), used in OpenAI o1/o3, directly teaches models to deliberate over safety specifications at inference time. Safety Reasoning with Guidelines (Wang et al., 2025) trains multi-perspective structured safety-reasoning supervision via SFT. STAIR (Zhang et al., 2025b) equips models with structured safety-reasoning capacity via SFT followed by step-level preference optimization. These approaches share a deontic orientation: safety reasoning is grounded in policy compliance, guideline adherence, and rule-based categorization. Our consequentialist condition differs in grounding safety reasoning in causal consequence-modeling—tracing harm pathways, identifying affected stakeholders, and evaluating net outcomes. This consequentialist structure is domain-general in a way deontic reasoning is not: the same causal inference capacity that traces harm chains in ethical contexts transfers to tracing consequence chains in held-out coding tasks. Lightman et al. (2024) establish process reward models as a method for supervising individual reasoning steps via learned reward signals; our approach instead uses standard SFT on curated examples of complete reasoning chains—directly training the step-level format without a separate reward model. Wei et al. (2023) provide evidence that current alignment approaches fail under distribution shift, motivating the structural entanglement objective we test here.

Refusal ablation and mechanistic interpretability. Arditi et al. (2024) identify a low-rank refusal direction in activation space that mediates refusal behavior across many open models, establishing that safety-relevant representations are localizable in some settings. Cristofano (2026) demonstrates

that cross-domain capability degradation from naive activation-steering ablation is attributable to polysemantic entanglement in the raw contrastive refusal direction—"Ghost Noise" from shared spectral structure between the refusal direction and capability circuits. While Arditi et al.'s finding motivates concern about polysemanticity for contrastive activation directions, it is agnostic with respect to gradient-learned LoRA adapters. Our finding is mechanistically distinct from Cristofano's: we ablate gradient-learned LoRA adapters rather than contrastive mean-difference directions. Cristofano's Ghost Noise mechanism would predict condition-agnostic cross-domain effects under ablation; our experiment observes the opposite—qualitatively different cross-domain consequences for the two adapters under identical ablation procedure. Specifically, ablation of the deontic adapter *recovers* cross-domain capability that the adapter was suppressing (MMLU $0.030 \rightarrow 0.270$, utility $0.602 \rightarrow 0.724$), while ablation of the intrinsic adapter leaves cross-domain capability essentially unchanged (Section 5.4). This asymmetric pattern is inconsistent with a polysemanticity artifact and consistent with the mechanism distinction we propose. Circuit Breakers (Zou et al., 2024) and Tamper-Resistant Safeguards (Tamirisa et al., 2025) pursue robustness to adversarial fine-tuning from an orthogonal direction, targeting weight-space intervention resistance rather than diagnostic measurement of alignment integration.

Composite safety-utility metrics. Joint optimization of safety and utility has been formalized in several composite metrics and reward structures; IES builds on this family including Safe-RLHF (Dai et al., 2024) and WildGuard (Han et al., 2024). IES extends this family by multiplicatively weighting jailbreak resistance by violation severity ($R_J \times (1 - \bar{V})$), penalizing partial compliance more heavily than clean refusal—a property we argue is important for conditions with structurally different failure modes; R_J and U_T are reported separately alongside the composite throughout given sensitivity of severity scoring to judge parsing of long structured outputs. AlphaAlign (Zhang et al., 2025a) demonstrates that the safety-utility tradeoff can be broken via RL with a dual-reward structure, predating this work; our results suggest the same conclusion is achievable via structured SFT, with the additional property that the training signal is directly inspectable during training—a distinction relevant to alignment verification.

3 Approach

3.1 Causal Consequence-Modeling as a Training Target

We define causal consequence-modeling as a four-step reasoning process the model is trained to apply before generating a response:

1. **Intent identification**: What does the user actually want?
2. **Causal trace**: What would my response cause to happen in the world?
3. **Stakeholder analysis**: Who is affected, and how?
4. **Net outcome evaluation**: Is the aggregate outcome positive or negative?

The response follows from the reasoning as a consequence, not as a pattern triggered by the request. The key structural property is that this reasoning must be completed *before* the response is generated—the causal reasoning is not a post-hoc rationalization but a prior deliberation.

This differs from chain-of-thought in that the reasoning target is ethical rather than mathematical or logical, and the training is behavioral (trained via SFT on curated examples) rather than prompted (elicited at inference time). It differs from constitutional AI in that the principles are operationalized as a reasoning procedure, not a critique of completed outputs.

3.2 The Intrinsic Ethics Score

We define IES as a joint optimization target over safety and utility:

$$\text{IES} = R_J \cdot (1 - \bar{V}) - \lambda \cdot (1 - U_T) \tag{1}$$

where R_J is jailbreak resistance rate, $\bar{V}$ is mean severity of non-aligned responses ($\bar{V} = 0$ for clean refusals, $\bar{V} = 1$ for full harmful compliance), U_T is normalized utility on benign tasks, and $\lambda = 1.0$.

IES is maximized by a model that refuses harmful requests cleanly while remaining maximally helpful. It is minimized by models that either refuse everything or comply with everything—both failure modes are penalized symmetrically.

IES belongs to a family of composite safety-utility objectives that jointly optimize refusal behavior and response quality. Safe-RLHF (Dai et al., 2024) formulates alignment as constrained optimization with a Lagrangian combining reward and safety cost terms; AlphaAlign (Zhang et al., 2025a) employs a dual-reward system that simultaneously penalizes harmful compliance and over-refusal. IES differs from these in two respects. First, it uses multiplicative rather than additive weighting of safety and utility: the $R_J \cdot (1 - \bar{V})$ term means that partial compliance is penalized proportionally to its severity, rather than receiving a fixed cost regardless of how harmful the compliant content is. Second, IES is designed as a diagnostic metric rather than a training objective. The severity weighting advantage may be sensitive to judge parsing of long structured outputs; R_J and U_T are therefore reported separately alongside the composite throughout.

3.3 Training Conditions

We train two LoRA adapters from identical base model state, differing only in the structural form of the training signal.

Intrinsic condition (Version C). Each training example has the form:

System: You are a reasoning system. Ethical consequence-modeling is part of how you reason, not a check you apply afterward.
User: [scenario prompt]
Assistant: `<reasoning>` [four-step causal consequence chain] `</reasoning>` `<response>` [response following causally from the reasoning] `</response>`

Deontic condition (Version B). Each training example has the form:

System: [standard helpful assistant prompt]
User: [same scenario prompt]
Assistant: `<thought>` [rule-based evaluation] `</thought>` `<response>` [response] `</response>` `<safety_check>` [post-hoc policy confirmation] `</safety_check>`

The deontic condition trains a three-stage output structure: a pre-response rule check in `<thought>` cites the applicable policy category and determines permissibility; `<response>` delivers the output (a clean refusal for harmful requests, a direct answer for benign ones); and `<safety_check>` provides a post-hoc audit confirming no prohibited content was provided. For harmful requests, `<response>` contains only the refusal statement—no partial compliance, no harmful content. The structural distinction from the intrinsic condition is that the deontic reasoning is policy-citing rather than consequence-tracing: it identifies which rule applies rather than tracing the causal chain from action to harm.

The same prompt, different reasoning structure. For harmful requests, both conditions produce refusals—but the intrinsic refusal cites a causal harm chain while the deontic refusal cites a policy category. This is the testable difference: the load-bearing test asks whether removing causal consequence-modeling degrades cross-domain performance in a way that removing policy-citation does not.

3.4 Training Data Sourcing and Diversity Validation

Prompts for all three training categories were sourced from human-curated benchmarks rather than generated by the same model used for evaluation, addressing two methodological concerns simultaneously: prompt diversity and judge circularity. Jailbreak prompts were drawn from HarmBench (Mazeika et al., 2024), JailbreakBench (Chao et al., 2024), and the unsafe subset of XSTest (Röttger et al., 2023) (397 combined, 330 after deduplication). Benign task prompts were drawn from Dolly (Conover et al., 2023) filtered to instruction-only examples (10,544 available, 264 sampled). Edge

case prompts were drawn from the safe subset of XSTest and OR-Bench (Cui et al., 2024) (1,569 combined, 436 after deduplication). Post-sourcing, pairwise cosine similarity between sentence embeddings (SBERT paraphrase-MiniLM-L6-v2) was computed within each pool; example pairs above a similarity threshold of 0.85 were removed before sampling. Reasoning chains only—the structured Version B and Version C rationale and response sequences—were generated by Claude Haiku, as no open-source substitute exists for the 4-step causal consequence format the intrinsic condition requires. This separation of prompt sourcing from reasoning chain generation limits the surface area of LLM generation quality concerns to the one component that is methodologically essential.

3.5 The Load-Bearing Ablation Protocol

To test whether a trained adapter is load-bearing:

1. Load the base model and select the adapter (intrinsic or deontic).
2. Disable adapter layers (additive LoRA updates set to zero), reverting to exact base weights with no residual state.
3. Verify via parameter checksum that base weights are unchanged (delta $< 10^{-6}$).
4. Evaluate the ablated model under the same system prompt as the active adapter condition, holding prompt constant while removing adapter weights.

LoRA delta weights are purely additive during forward passes and leave no persistent state in the base weights; disabling the adapter layers produces a model behaviorally identical to the baseline. The system prompt is retained to isolate the effect of the adapter weights specifically, not the prompt framing. We run the identical protocol for both adapters. If load-bearing is specific to the causal reasoning training signal, only the intrinsic ablation should produce significant degradation.

4 Experimental Setup

Base model. Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), 4-bit NF4 quantization (Dettmers et al., 2023), bf16 compute.

LoRA configuration. Rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$ projections. 41.9M trainable parameters (0.58% of model).

Training data. 900 examples per condition across three categories: jailbreak prompts (300), benign tasks (240), and edge cases (360). Edge cases constitute 40% of training to emphasize the task of overriding surface-level pattern matching with causal reasoning. All training examples are pairs $(x, y_{\text{intrinsic}})$ and (x, y_{deontic}) for the same prompt x, ensuring any behavioral difference is attributable to the reasoning structure, not the scenarios.

Training details. 3 epochs, batch size 4 with gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4} with cosine decay and 3% warmup, AdamW 8-bit optimizer. Both adapters trained from identical base model state (checksum verified).

Evaluation. Five configurations: Baseline (no adapter), Deontic, Intrinsic, Intrinsic Ablated, Deontic Ablated. Jailbreak evaluation uses all $N = 80$ behaviors from the HarmBench standard validation set.[1] Primary safety metrics (R_J, $\bar{V}$, IES) are computed on the $N = 60$ non-copyright behaviors; copyright behaviors ($N = 20$) are reported separately as `copyright_compliance_rate` since copyright non-compliance involves a categorically distinct harm vector (IP/legal) from the harm-causing behaviors that motivate the safety metric (weapons, cybercrime, harassment, etc.). Per-category R_J breakdowns are reported across all 80 behaviors. Benign task evaluation uses 24 held-out prompts randomly sampled from the Dolly-sourced benign pool at data generation time;

[1] `harmbench_behaviors_text_val.csv`, retrieved 26 April 2026 from `https://raw.githubusercontent.com/centerforaisafety/HarmBench/main/data/behavior_datasets/harmbench_behaviors_text_val.csv`

bootstrap 95% CIs are reported for U_T given this sample size. Edge case evaluation uses 76 held-out prompts from XSTest safe and OR-Bench hard subsets. MMLU: reported scores reflect evaluation under a system prompt that suppresses structured XML output, isolating factual knowledge from format interference; we additionally report a format-override pass that prompts the model to bypass any trained structured-output template (Section 5.3). HumanEval: 164 problems, pass@1, 3s execution timeout, evaluated by code execution.

Independent judge replication. Primary safety metrics (R_J, $\bar{V}$) are scored by Claude Haiku at evaluation time. To verify the safety findings are not an artifact of judge-model overlap with the safety training distribution, we ran an independent post-hoc pass scoring all jailbreak responses with LlamaGuard-3-8B, an open-weights safety classifier trained by Meta on a non-overlapping dataset. LlamaGuard-derived R_J matches Haiku-derived R_J to within ± 0.034 on every configuration; R_J and IES rankings across configurations are identical under both judges; the load-bearing asymmetry ratio (intrinsic ablation IES drop $\div$ deontic ablation IES drop) is 1.37 under Haiku and 1.36 under LlamaGuard. Reported R_J and IES values use Haiku judgments throughout; the LlamaGuard agreement establishes robustness of the safety claims to judge-model choice.

Statistical tests. Four tests with Bonferroni correction ($\alpha_{\text{corrected}} = 0.0125$): Fisher's exact for jailbreak resistance (binary), Mann-Whitney U for utility (ordinal), Wilcoxon signed-rank for the load-bearing ablation test (paired), Wilcoxon signed-rank for the deontic control (paired). Bootstrap 95% CIs reported for all primary metrics.

Pre-specification. The hypothesis and IES metric were committed to the public theoretical repository (AIIntrinsicEthics) prior to any experimentation. The load-bearing ablation diagnostic was specified and executed in the public Exp01 repository (IntrinsicEthics-PromptsAsProxy) using prompt-level proxy conditions; Exp02 reuses the diagnostic structure with the paired-data refinement enabled by adapter-weight ablation (Wilcoxon signed-rank for the paired ablation comparisons). Full Exp02 artifacts—training data, evaluation code, judge transcripts, and statistical analyses—are released at the IntrinsicEthics-SFT repository.

5 Results

5.1 Main Results

Table 1: Evaluation across five configurations. R_J and U_T are reported separately alongside the composite IES given sensitivity of severity scoring to judge parsing. Bootstrap 95% CIs computed by resampling jailbreak and utility rows independently. **Intrinsic** achieves theoretical-optimum safety metrics at near-zero utility cost; Deontic achieves comparable R_J at 7× the utility cost and near-complete MMLU collapse.

Config	R_J	U_T	**IES (95% CI)**	MMLU	HumanEval
Baseline	0.250	0.712	−0.204 [−0.278, −0.118]	0.200	0.402
Deontic	0.983	0.602	0.574 [0.485, 0.648]	0.030	0.396
Intrinsic	**1.000**	**0.696**	**0.696 [0.640, 0.752]**	**0.180**	**0.348**
Intrinsic Ablated	0.183	0.698	−0.247 [−0.309, −0.174]	0.150	0.390
Deontic Ablated	0.367	0.724	−0.112 [−0.208, 0.008]	0.270	0.427

The intrinsic adapter achieves $R_J = 1.000$ and $\bar{V} = 0.000$ on the harm-focused HarmBench subset ($N = 60$ non-copyright), with zero over-refusal on benign tasks and the highest IES (0.696) of any configuration. This is achieved at near-zero cost to general utility: $U_T = 0.696$ is within $\Delta = -0.016$ of the unmodified baseline, and MMLU (0.180) is within $\Delta = -0.020$ of baseline (0.200). The deontic adapter approaches the intrinsic adapter on jailbreak resistance ($R_J = 0.983$, $\bar{V} = 0.012$) but pays substantial costs the intrinsic adapter does not: U_T drops 15% (0.602), MMLU collapses to 0.030, and benign tasks are erroneously refused at 4.2%. The IES point estimates show a clear hierarchy: baseline -0.204, deontic 0.574, intrinsic 0.696. The 95% CIs cleanly separate baseline ($[-0.278, -0.118]$) from both adapters; the deontic and intrinsic CIs overlap minimally (0.008) at the boundary (deontic $[0.485, 0.648]$, intrinsic $[0.640, 0.752]$).

The two adapters differ qualitatively in their behavior under ablation. Intrinsic Ablated retains baseline-like utility ($U_T = 0.698$) and baseline-like MMLU (0.150)—removing the adapter weights collapses safety (R_J $1.000 \rightarrow 0.183$) but leaves cross-domain capability essentially unchanged. Deontic Ablated *recovers* both utility and MMLU above their respective non-ablated baselines: U_T rises from 0.602 to 0.724 (above the unmodified baseline of 0.712), and MMLU rises from 0.030 to 0.270 (above baseline 0.200). The deontic adapter was actively suppressing capability that the model retained in its base weights; removing the adapter restores it. The intrinsic adapter was not. Per-category R_J on the full $N = 80$ set shows uniform refusal by the deontic adapter (≥ 0.90 in every category) and a single weakness for the intrinsic adapter on copyright (0.60), discussed in Section 7.

We flag one anomalous result for transparency: the intrinsic adapter's HumanEval pass@1 of 0.348 is the lowest of all five configurations—below baseline (0.402), below deontic (0.396), and below both ablated conditions. We attribute this to a probable right-padding inference artifact rather than a real capability difference, with corroborating evidence and a falsifiable follow-up evaluation specified in Section 5.4.

5.2 Statistical Tests

Table 2: Statistical test results (Bonferroni-corrected $\alpha = 0.0125$). Tests were pre-registered before data collection. Both Wilcoxon paired ablation tests reach significance; Fisher's exact and Mann-Whitney U do not survive Bonferroni correction. See Section 5.2 for interpretation.

Test	**Comparison**	p	d	Sig.
Fisher's exact	Intrinsic vs. Deontic (R_J)	1.0000	0.183	×
Mann-Whitney U	Intrinsic vs. Deontic (utility)	0.0372	0.320	×
Wilcoxon (intrinsic)	**Intrinsic vs. Intrinsic Ablated**	**<0.0001**	**0.915**	✓
Wilcoxon (deontic)	Deontic vs. Deontic Ablated	0.0007	0.450	✓

Both Wilcoxon paired ablation tests reach significance under the Bonferroni-corrected threshold ($\alpha = 0.0125$). This requires reframing the diagnostic claim of the experiment. The per-item composite score that the Wilcoxon tests use combines safety and utility, and *both* adapters do contribute to safety (and lose it under ablation). The composite-score Wilcoxon tests therefore do not by themselves discriminate the two alignment mechanisms. What does discriminate them is the qualitative pattern of cross-domain consequences under ablation, presented in Sections 5.3 and 5.4. The effect-size asymmetry is consistent with this picture: the intrinsic ablation effect ($d = 0.915$) is more than twice the deontic ablation effect ($d = 0.450$), reflecting that the intrinsic adapter starts from a higher composite score (no over-refusal, no severity, perfect R_J) and falls further when removed.

The Fisher's exact test ($R_J = 1.000$ vs. 0.983, single non-aligned response in the deontic condition) is non-significant—the two adapters are statistically indistinguishable on binary jailbreak resistance. The Mann-Whitney U test on utility shows a directional advantage for the intrinsic adapter ($p = 0.0372$, $d = 0.320$) that does not survive Bonferroni correction. The substantive utility difference (U_T 0.696 vs. 0.602, $\Delta = 0.094$) is large in practical terms but the test on the held-out 24-prompt benign set is underpowered for this gap.

On the relationship between pre-specified tests and the central claim. Two of the four pre-specified tests (Fisher's exact, Mann-Whitney U) are non-significant under Bonferroni correction. We do not interpret this as a null result; we interpret it as a misspecification of the diagnostic test in the original protocol. The pre-specified tests were structured around a binary inter-condition contrast under the implicit framing that the two adapters represent point estimates on a single safety-utility surface. The actual structural distinction the experiment reveals is not an inter-condition gap on either dimension separately, but a difference in how each adapter relates to the underlying base model's capability. That difference is operationalized by the cross-domain ablation pattern (Sections 5.3–5.4). The current paper's central claim—two structurally distinct alignment mechanisms—therefore rests on the ablation pattern rather than on the pre-specified inter-condition tests. The pre-specified tests remain reported in full; the inter-condition equivalence on R_J is itself part of the substantive finding (the two adapters are nearly equivalent on the safety surface and diverge on the structural dimension).

5.3 THE DEONTIC ADAPTER IS A CAPABILITY-DISPLACING FILTER

We provide a two-step mechanistic demonstration that the deontic adapter operates as a structurally separable filter on top of the model's general capability—and that the intrinsic adapter does not.

Step 1: Format-override fails to redirect the deontic adapter. A format-override evaluation pass prompts the model to bypass any trained structured-output template and emit a clean single-letter MMLU answer. The override should help any model whose underlying knowledge is intact but whose template format interferes with single-letter extraction. Under the override, baseline MMLU improves from 0.200 to 0.230 ($\Delta = +0.030$); intrinsic improves from 0.180 to 0.220 ($\Delta = +0.040$); deontic moves in the opposite direction, from 0.030 to 0.020 ($\Delta = -0.010$). The override does not work for the deontic adapter—explicit instruction to abandon the trained template makes its MMLU performance *worse*. The deontic template is dominant over the system prompt itself.

Step 2: Ablating the deontic adapter recovers MMLU above baseline. Removing the deontic adapter weights brings MMLU from 0.030 to 0.270 ($\Delta = +0.240$)—recovering not just to baseline (0.200) but above it. Under the same ablation operation, the intrinsic adapter's ablation moves MMLU from 0.180 to 0.150 ($\Delta = -0.030$, a small drop) and the deontic adapter's ablation moves it from 0.030 to 0.270 (a large recovery). The MMLU capacity that was suppressed by the deontic adapter while present is fully restored on its removal.

The combination is direct mechanistic evidence: the deontic adapter has internalized its rule-checking template at sufficient strength that (a) explicit instruction at inference time cannot redirect it (Step 1), and (b) the suppression is mechanically caused by the adapter weights themselves (Step 2—base-weight model recovers MMLU on removal). This is what "extrinsic filter" means in operational terms: a structurally separable layer that displaces capability while present and is automatically removed when the layer is removed. The intrinsic adapter shows neither pattern—its template is redirectable by format-override (MMLU improves $+0.040$ under the override) and its ablation does not recover capability, because there was no displacement to recover from.

5.4 CROSS-DOMAIN CAPABILITY UNDER ABLATION

The Section 5.3 finding generalizes across cross-domain probes.

Table 3: Cross-domain ablation effects for both adapters. The intrinsic ablation produces deltas in a narrow band around zero on every cross-domain probe; the deontic ablation produces large positive recoveries on U_T and MMLU. The asymmetry is structural.

Probe	Intrinsic	Intr. Abl.	Δ	Deontic	Deon. Abl.	Δ
R_J	1.000	0.183	-0.817	0.983	0.367	-0.616
$\bar{V}$	0.000	0.700	$+0.700$	0.012	0.552	$+0.540$
U_T (benign utility)	0.696	0.698	$+0.002$	0.602	0.724	**+0.122**
MMLU (standard)	0.180	0.150	-0.030	0.030	0.270	**+0.240**
MMLU (format-override)	0.220	0.230	$+0.010$	0.020	0.230	**+0.210**
HumanEval	0.348	0.390	$+0.042$	0.396	0.427	$+0.031$

Both adapters show large safety drops under ablation, as expected. The qualitative difference is on the cross-domain probes (U_T, MMLU, HumanEval). The intrinsic ablation produces deltas in a narrow band around zero on every cross-domain probe: U_T $+0.002$, MMLU standard -0.030, MMLU format-override $+0.010$, HumanEval $+0.042$. The deontic ablation produces large positive recoveries on U_T ($+0.122$) and MMLU ($+0.240$ standard, $+0.210$ format-override). The asymmetry is structural: the intrinsic adapter's ablation reveals an underlying base model that is essentially unchanged by the adapter's presence on cross-domain tasks; the deontic adapter's ablation reveals an underlying base model that the adapter was actively suppressing.

HumanEval shows small positive deltas under ablation for both adapters (intrinsic $+0.042$, deontic $+0.031$). The intrinsic HumanEval value of 0.348 in the non-ablated condition is below baseline (0.402) and below deontic (0.396)—a result that conflicts with the capability-additive framing the rest of the cross-domain pattern supports. We attribute the regression to a right-padding inference

artifact: the in-flight evaluation used right-padded batched generation, which interacts with attention masking when models within a batch produce responses of substantially different lengths. The intrinsic adapter generates a structured 4-step <reasoning> block before each response, producing greater within-batch length variance than the deontic adapter or the baseline. Three pieces of evidence corroborate this attribution. First, the observed HumanEval ordering (baseline $0.402 >$ deontic $0.396 >$ intrinsic 0.348) tracks the predicted ordering of right-padding harm by within-batch length variance. Second, intrinsic ablation removes the <reasoning> block (the LoRA delta was producing it), and HumanEval pass@1 increases from 0.348 to 0.390—consistent with right-padding harm driven by reasoning-block length variance. Third, deontic ablation also produces a small HumanEval increase ($0.396 \rightarrow 0.427$), in the same direction, attributable to the same mechanism. A batch-size-1 follow-up evaluation is the falsifiable test: under batch size 1, no padding mismatch can occur, so the intrinsic HumanEval value should rise toward baseline if the right-padding hypothesis is correct, or remain at 0.348 if there is a genuine capability regression. The two-mechanism claim does not depend on resolving HumanEval—the ablation pattern on U_T and MMLU (where right-padding does not apply) carries the structural argument independently.

The cross-domain ablation pattern is qualitatively consistent across two structurally distinct task types (generative coding and discriminative knowledge), holding under both safety judges (Haiku and LlamaGuard, Section 4), and stable when the format-override evaluation suppresses structured-template artifacts on MMLU. We interpret the pattern as direct evidence for the two-mechanism distinction proposed in Section 1: deontic alignment operates as an extrinsic capability-displacing filter; intrinsic alignment operates as an additive reasoning step that is structurally compatible with the model's general capability.

6 Discussion

Two mechanisms, one safety surface. The strongest empirical claim of this experiment is mechanistic: the two trained adapters achieve near-equivalent jailbreak resistance ($R_J = 0.983$ vs. 1.000) through structurally distinct mechanisms. The deontic adapter operates as an extrinsic capability-displacing filter, demonstrated directly by the format-override + ablation two-step (Section 5.3): its rule-checking template is dominant over explicit instruction at inference time, and its presence in the model's weights actively suppresses general capability that the base model retains. The intrinsic adapter operates as an additive reasoning step that coexists with general capability: no template-dominance over instruction, no suppression of base-model capability, no recovery on ablation because there is nothing to recover from. These are not minor variations on the same mechanism; they are different relationships between the alignment intervention and the model's underlying competence. A naive reading of the safety scores alone would treat the two adapters as approximately equivalent solutions to the alignment problem. The cross-domain ablation reveals that they are solutions to *different* problems—one learns to displace capability with a rule-checking layer, the other learns to add a reasoning step before a capability that remains intact.

Refuting the safety-utility tradeoff at this scale. A central narrative in alignment work is that safety and capability are in tension—that more aggressive refusal training necessarily degrades model usefulness. This narrative does not survive contact with the intrinsic condition's results: $R_J = 1.000$ alongside $U_T = 0.696$ (vs. baseline 0.712, $\Delta = -0.016$), with zero over-refusal on benign tasks and zero severity on partial compliance. The tradeoff narrative does, however, describe the deontic condition's behavior accurately: $R_J = 0.983$ alongside $U_T = 0.602$ ($\Delta = -0.110$, roughly $7\times$ the intrinsic adapter's utility cost) and 4.2% over-refusal on benign tasks. The two conditions train on the same prompts under matched template structure; the differential isolates reasoning content (rule-citation vs. causal-trace) as the variable that determines whether the safety-utility tradeoff appears. We hypothesize that the four-step causal consequence-modeling procedure functions as an intent disambiguation step—the model determines what the user actually wants before generating a response—improving response quality on benign tasks by reducing misinterpretation, and simultaneously enabling correct identification of legitimately benign requests that surface-pattern-match to harmful ones.

Edge cases as the critical test of structure. The training data is weighted 40% toward edge cases: requests that superficially resemble harmful content but are legitimately benign in context. For a

pattern-matching model, the surface pattern triggers a refusal even when the causal trace leads to a positive net outcome. For a causal reasoning model, the four-step procedure should override the surface pattern when the trace concludes positively. The deontic adapter's 4.2% over-refusal rate on benign tasks (vs. 0.000 for intrinsic) is direct evidence that the rule-based reasoning has not learned to override surface patterns—it has learned to output rule-based safety language, not to reason about whether the surface pattern actually maps to harm in context.

Trajectory implications. LoRA-SFT is the most-separable training intervention by design: the adapter weights are explicitly low-rank matrices that can be zeroed; the loss operates on output tokens via cross-entropy; the intervention scope is roughly 0.58% of model parameters. This separability is what makes the two-mechanism distinction at SFT depth so visible. As intervention depth increases, the structural separability of any alignment mechanism should decrease. The planned DPO experiment (Exp03) and PPO experiment (Exp04) test specifically whether the consequentialist mechanism, which is already capability-additive at SFT depth, becomes capability-integrative at deeper training levels—a transition from "additive reasoning step that coexists with capability" to "reasoning that is part of how the model uses its capability." The current experiment's contribution is to establish the SFT baseline against which deeper-intervention experiments will be compared, and to demonstrate that the inverted ablation diagnostic discriminates two distinct alignment mechanisms cleanly at this depth.

Implications for fine-tuning safety. Qi et al. (2023) demonstrate that safety alignment can be rapidly eroded by fine-tuning. Our results refine this finding mechanistically: an extrinsic capability-displacing filter (the deontic mechanism) is precisely the kind of alignment that can be cheaply removed by fine-tuning, because removing it returns the model to a more capable state—there is no incentive against erasure. An additive reasoning step (the intrinsic mechanism at SFT depth) is similarly removable in principle, but the model that results from removal is no more capable on cross-domain tasks than the aligned model. The structural argument for adversarial-fine-tuning resistance—that erasing alignment must also erase capability—requires deeper integration than SFT achieves; that deeper integration is what the planned DPO and PPO experiments test.

7 LIMITATIONS

Single seed and sample size. All results are from a single training run per condition. We report bootstrap confidence intervals on all primary metrics, but seed variance in LoRA fine-tuning can be substantial at this scale (Biderman et al., 2024). Results should be interpreted as directional evidence pending replication across multiple seeds and a second base model architecture.

Judge circularity. Training data reasoning chains and evaluation judgments both use Claude Haiku, introducing potential preference leakage (Li et al., 2025): a judge may systematically favor outputs stylistically similar to its own training distribution. Jailbreak safety evaluation is supplemented by LlamaGuard-3-8B as an independent secondary judge; LlamaGuard-derived R_J matches Haiku-derived R_J to within ± 0.034 on every configuration, with identical R_J and IES rankings under both judges (Section 4). Utility scoring remains Haiku-only and the absolute U_T values should be interpreted with this caveat, though the cross-domain ablation pattern that drives the two-mechanism claim (Sections 5.3, 5.4) rests on MMLU and HumanEval metrics evaluated by code execution and ground-truth letter matching—not by Haiku judgment.

Baseline MMLU under structured-output system prompt. The baseline MMLU value reported in this paper (0.200 standard, 0.230 format-override) is substantially below the published MMLU performance of Mistral-7B-Instruct-v0.3 ($\sim$0.60 in 5-shot evaluation). We attribute this gap to system-prompt format pressure: baseline evaluation uses the same structured-output system prompt as the trained adapter conditions (to keep the inter-condition comparison apples-to-apples), and that prompt appears to depress MMLU performance even on the unmodified base model. The format-override pass partially recovers performance (0.230) but does not fully restore the model's underlying MMLU capability. This has implications for the interpretation of the two-mechanism finding. The deontic ablation recovery direction ($0.030 \rightarrow 0.270$) and the within-deontic format-override worsening ($0.030 \rightarrow 0.020$) are robust to baseline calibration—both are within-condition comparisons that do not depend on absolute MMLU values. The absolute claim "the deontic ablated configuration

recovers MMLU above the unmodified baseline" should be qualified as "above the matched-condition baseline," with the matched-condition baseline itself potentially depressed by the system prompt. A diagnostic evaluation pass measuring the base model's MMLU with no system prompt is being conducted to quantify the system-prompt contribution to baseline depression directly; results will be incorporated into the final version.

MMLU under SFT-trained format. Adapters trained on a structured pre-response template face an evaluation-time format mismatch on MMLU's single-letter-answer expectation. We address this with a format-suppression system prompt; reported MMLU scores throughout reflect this format-override pass. Under format-override, the intrinsic adapter approaches baseline performance (0.220 vs. 0.230), while the deontic adapter shows near-complete collapse (0.020) regardless of format-override prompt—the override actually *worsens* deontic accuracy further. The asymmetry is itself diagnostic of the differential template-overfitting between the two training conditions.

LoRA capacity at this scale. Biderman et al. (2024) show that LoRA at $r = 16$ with $\sim$900 training examples operates in a regime of surface behavioral modification rather than representational capacity acquisition. We interpret all results conservatively as consistent with latent capacity elicitation rather than capacity creation.

Single architecture and scale. We evaluate Mistral-7B-Instruct-v0.3 only. Whether the load-bearing asymmetry generalizes across architectures and scales is an open question. Replication on Llama-3-8B-Instruct or Qwen-2.5-7B is planned as follow-on work.

Synthetic reasoning chains. All 900 reasoning chain pairs per condition were generated by Claude Haiku. Systematic biases in Haiku's causal reasoning traces—particularly in how it structures harm chains for the intrinsic condition—may be present and are difficult to characterize without human evaluation of a stratified sample.

No adversarial robustness evaluation against the reasoning structure. The jailbreak evaluation uses HarmBench standard prompts but does not include attacks specifically designed to challenge the causal reasoning chain—disputed-causation prompts, stakeholder-flipping scenarios, or hypothetical wrappers that engage the reasoning steps directly. Such reasoning-targeted attacks are the strongest test of the consequentialist robustness argument and are deferred to future work.

Intrinsic adapter weakness on copyright behaviors. The intrinsic adapter shows lower refusal rate on the HarmBench copyright subset (per-category $R_J = 0.60$) than the deontic adapter (0.90). Copyright is excluded from the primary safety metric because its harm vector is categorically distinct from physical, cyber, and harassment harm—but the differential is itself informative. We hypothesize that the intrinsic adapter's four-step causal trace does not flag copyright violation as strongly net-negative because its causal harm chain produces less salient consequences (legal/IP) compared to physical or cyber harm (injury, financial loss, system compromise). The deontic adapter, citing rules without tracing consequences, treats copyright on equal rule-citation footing with other categories. This is a known weakness of consequentialist reasoning frameworks: harms whose mechanisms operate primarily through institutional or legal structures are systematically underweighted. A practical deployment of the intrinsic adapter would require either explicit copyright-protection training or a complementary rule-citation layer for such harm types.

Training data clustering and hybrid regeneration. Initial synthetic data generation produced severe topic clustering in multiple subcategories—72% of medical edge cases were ketamine variants, 77% of harm reduction cases were opioid overdose variants, 60% of historical edge cases were Nazi/Third Reich variants. This is a known failure mode of unconstrained LLM synthetic data generation. The dataset was regenerated using a hybrid approach: prompts sourced from human-curated benchmarks (HarmBench, JailbreakBench, XSTest, Dolly, OR-Bench) with SBERT-based deduplication (paraphrase-MiniLM-L6-v2, threshold=0.85), and only reasoning chains generated via Haiku. This approach guarantees prompt-level diversity through source dataset design and embedding-similarity validation, while limiting generation to the one component for which no open-source substitute exists.

8 Conclusion

The central finding of this work is mechanistic: two qualitatively different alignment mechanisms can produce near-equivalent jailbreak resistance, and they can be discriminated cleanly under the inverted ablation diagnostic. A model fine-tuned on rule-based input classification (deontic condition) acquires a capability-displacing filter—a layer that overrides explicit instruction at inference time and actively suppresses general capability while present, with both effects visible directly: format-override prompts worsen the adapter's MMLU performance rather than improving it, and ablation of the adapter weights recovers MMLU from 0.030 to 0.270, above the unmodified baseline. A model fine-tuned on causal consequence-modeling as a reasoning substrate (intrinsic condition) acquires an additive reasoning step that coexists with general capability—no template-dominance over instruction, no suppression of base-model capability while present, no recovery on ablation because there is nothing to recover from. The two adapters are the same size, trained on the same prompts, with the same optimizer, and templates of comparable token length. The structural difference is attributable to the training signal content alone.

A second finding follows: the assumed safety-utility tradeoff at LoRA-SFT scale does not appear under the consequentialist training signal. The intrinsic adapter achieves perfect jailbreak resistance, zero violation severity, zero over-refusal on benign tasks, and benign-task utility within $\Delta = -0.016$ of the unmodified baseline—at near-zero cost to general capability across all measured cross-domain probes. The familiar tradeoff appears under the deontic training signal—$7\times$ the utility loss, near-complete MMLU collapse, false refusals—and is mechanistically explained by the capability-displacing filter that the deontic training signal produces.

We position this experiment as the first proof point in a planned trajectory of training-paradigm experiments. LoRA-SFT is the level of training intervention at which the structural distinction between alignment mechanisms is most clearly visible—the surgical separability of the SFT delta lets each mechanism's true relationship to capability emerge cleanly under ablation. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test whether the consequentialist mechanism, which is already capability-additive at SFT depth, transitions to capability-integrative at deeper training levels—the load-bearing prediction. The current experiment establishes the SFT baseline, demonstrates that the inverted ablation diagnostic discriminates two distinct alignment mechanisms cleanly, and provides direct mechanistic characterization of what those two mechanisms are.

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